Entry Name: **JOOS**
VAST Challenge 2025
Challenge 3

Team Members:
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Student Team: Yes

Tools Used:
Docker, Gitlab, Kubernetes
Frontend: Next JS + React, D3, Three.js, Sigma.js, HeroUi
Backend: FastAPI, Ollama, Python , Jupyter
Database: Neo4J

Approximately how many hours were spent working on this submission in total?
360 hours

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2025 is complete? Yes
Video
https://cloud.uni-konstanz.de/index.php/s/a4jWKXSfPsSHZRj

Use visual analytics to analyze the available data and develop responses to the questions to be provided. In addition, prepare a video that shows how you used visual analytics to solve this challenge.

Note: the VAST challenge is focused on visual analytics and graphical figures should be included with your response to each question. Please include a reasonable number of figures for each question (no more than about 6) and keep written responses as brief as possible (around 250 words per question).

Questions

Clepper diligently recorded all intercepted radio communications over the last two weeks. With the help of his intern, they have analyzed their content to identify important events and relationships between key players. The result is a knowledge graph describing the last two weeks on Oceanus. Clepper and his intern have spent a large amount of time generating this knowledge graph, and they would now like some assistance using it to answer the following questions.

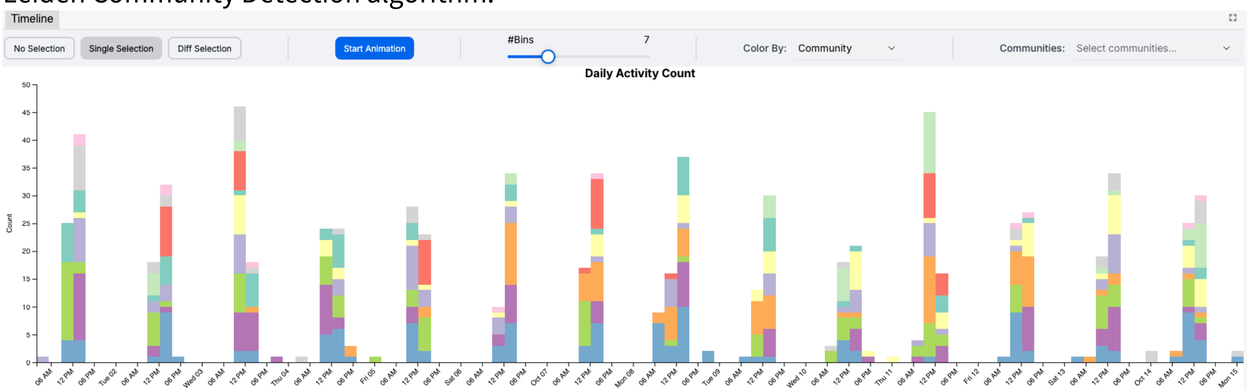
1 – Clepper found that messages frequently came in at around the same time each day.

a. Develop a graph-based visual analytics approach to identify any daily temporal patterns in communications.
b. How do these patterns shift over the two weeks of observations?
c. Focus on a specific entity and use this information to determine who has influence over them.

1a) Identifying Daily Temporal Patterns

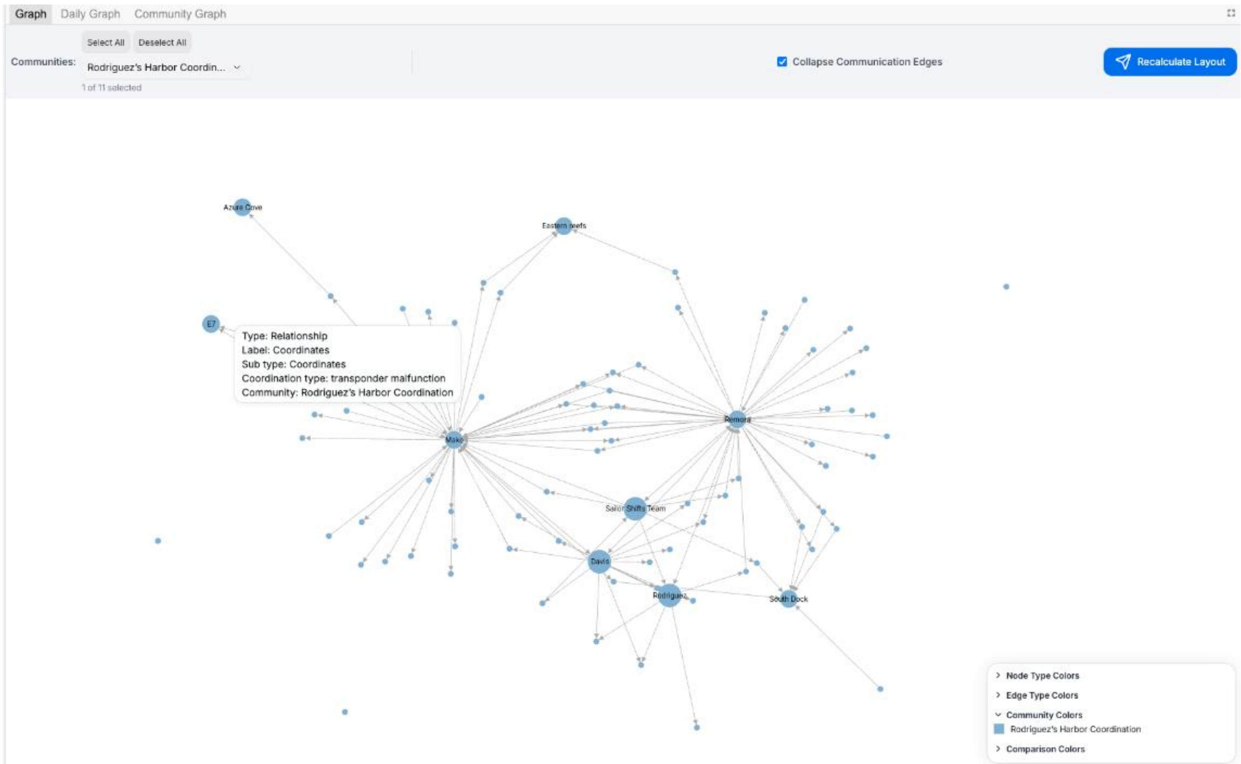
To identify daily temporal patterns in communications, we used a timeline component with brushing, linked to the knowledge graph. This allows us to select a time range and immediately see which events and entities are active during that period.

The following visualization shows the overall temporal pattern by community, as detected using the Leiden Community Detection algorithm:

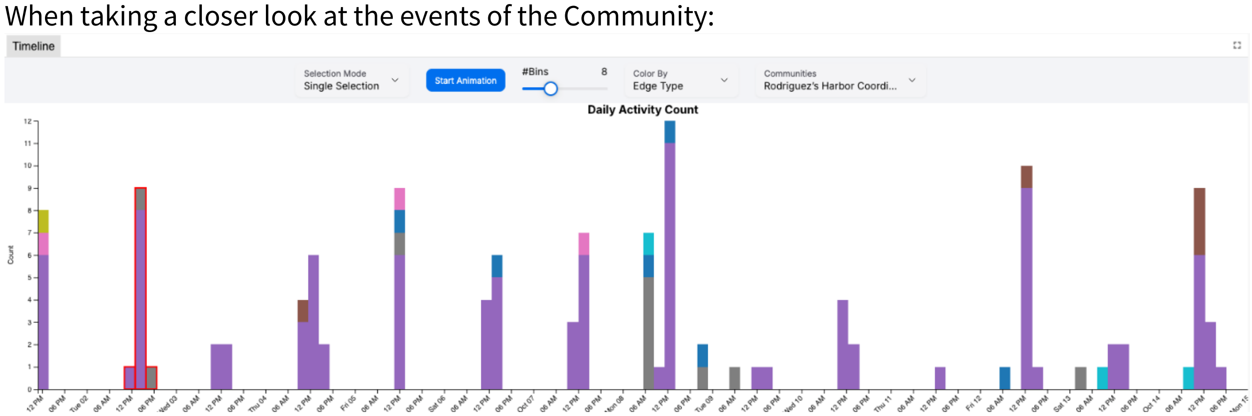


From this visualization we observe that most of the events take place between 10:00 AM and 04:00 PM, distributed between the different communities. There is a consistent peak in activity between 10:00 AM and 12:00 PM each day.

The community, which is present every day is the Rodriguez Harbor Coordination (Blue Community) with its Entities displayed in this graph:

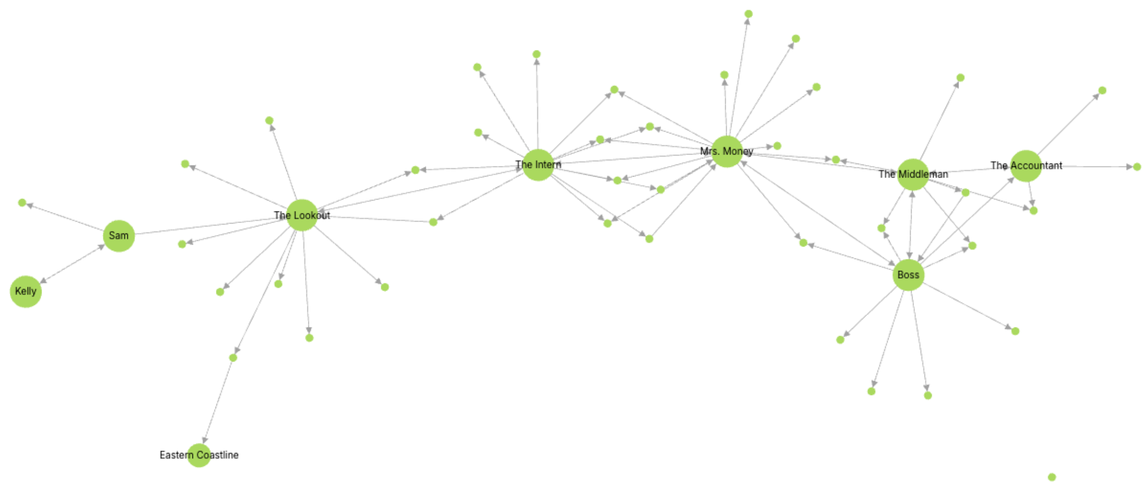


When taking a closer look at the events of the Community:



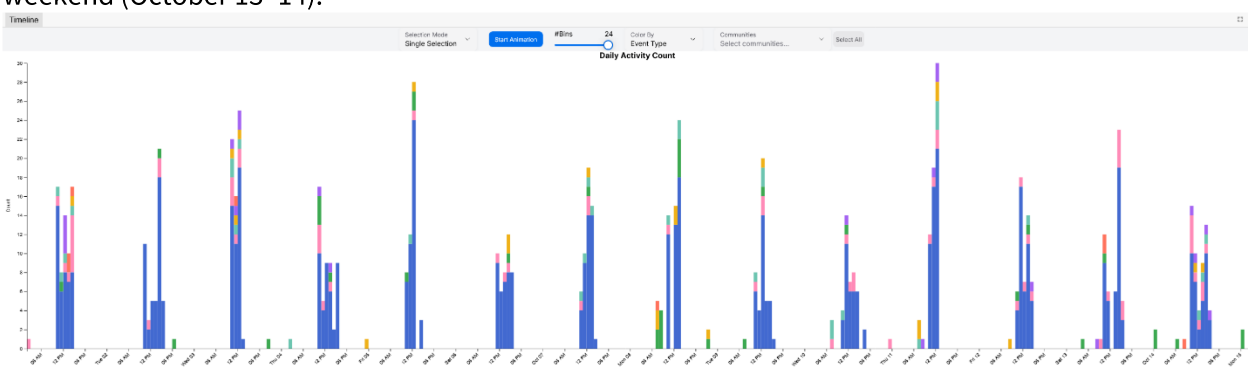
It is visible at they are mostly communicating (purple bars), with some vessel activity, enforcements, assessments, collaborations and tour activities in between.

The second most present community is the Nemo Reef Operations community (Light Green Community), which has events always around 12AM, except on 06.10.



1b) Shifts in Patterns Over Two Weeks

Examining the temporal patterns over the two-week period reveals the following:
Intercepted communications consistently occur daily from 11:00 AM to 04:00 PM, including both weekdays and weekends.
There is noticeably less activity during the first weekend (October 6–7) compared to the second weekend (October 13–14).



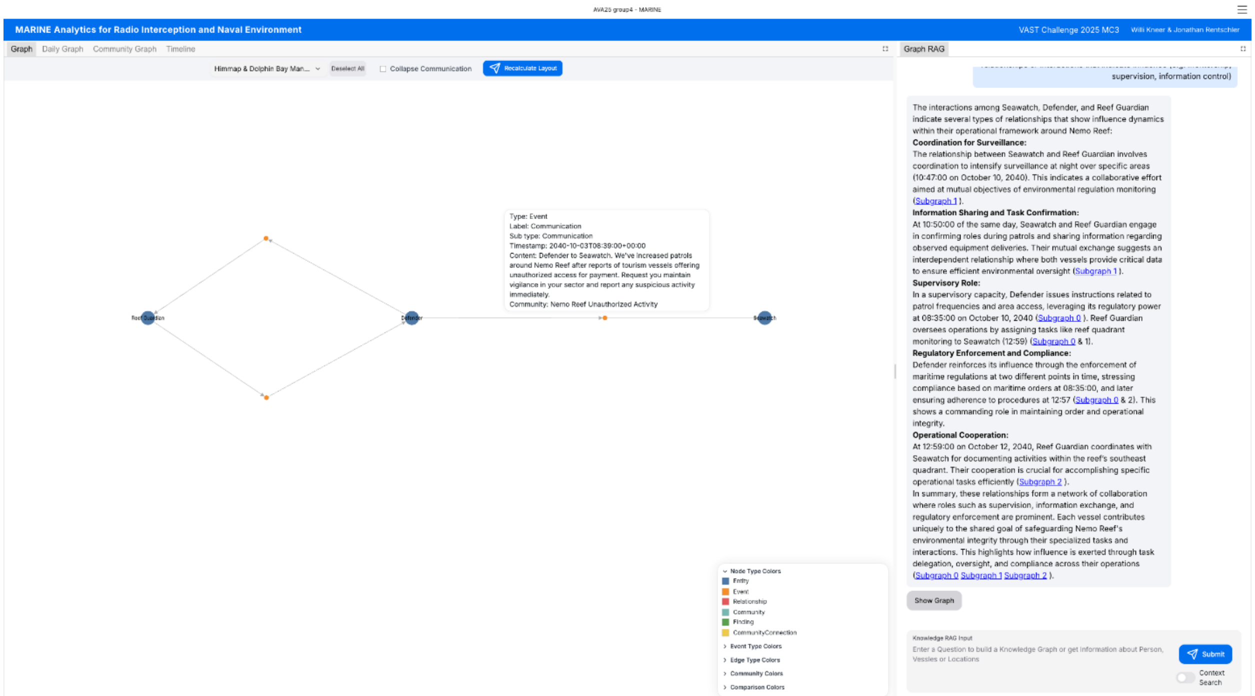
This suggests that while communication is generally steady, there are fluctuations in activity levels, particularly on weekends.

1c) Entity Focus:

We can determine who has influence over a specific entity by sending the following prompt to the GraphRAG pipeline:

> *I observed that the vessels Seawatch, Defender and Reef Guardian are closely associated with each other. Determine the types of relationships or interactions that indicate influence (e.g. mentorship, supervision, information control)*

Output:



The model identifies the relationship between Defender and Seawatch as a supervisory kind with, Defender issuing instructions related to patrol frequencies and area access.

2 – Clepper has noticed that people often communicate with (or about) the same people or vessels, and that grouping them together may help with the investigation.

- **a.** Use visual analytics to help Clepper understand and explore the interactions and relationships between vessels and people in the knowledge graph.
- **b.** Are there groups that are more closely associated? If so, what are the topic areas that are predominant for each group?
    - For example, these groupings could be related to: Environmentalism (known associates of Green Guardians), Sailor Shift, and fishing/leisure vessels.

2a) Visual Analytics for Exploring Interactions and Relationships

To help Clepper explore the interactions and relationships between vessels and people, we applied the Leiden algorithm to detect communities within the knowledge graph. This approach revealed eleven distinct communities, with each node assigned to a specific group.

In our Graph RAG pipeline, these communities are automatically detected and summarized using a large language model. The summaries are accessible by hovering over nodes in the `Community Graph` visualization, providing instant context for each group.

The identified communities are as follows:
1. Nemo Reef Protection
2. Himmap & Dolphin Bay Management
3. Rodriguez's Harbor Coordination
4. Nemo Reef Operations
5. Restricted Access Issues
6. Nemo Reef Monitoring by EcoVigil
7. Nemo Reef operations by V. Miesel
8. Access Management
9. Nemo Reef interactions by Oceanus City
10. Nemo Reef Unauthorized Activity
11. Nemo Reef & Haacklee Harbor

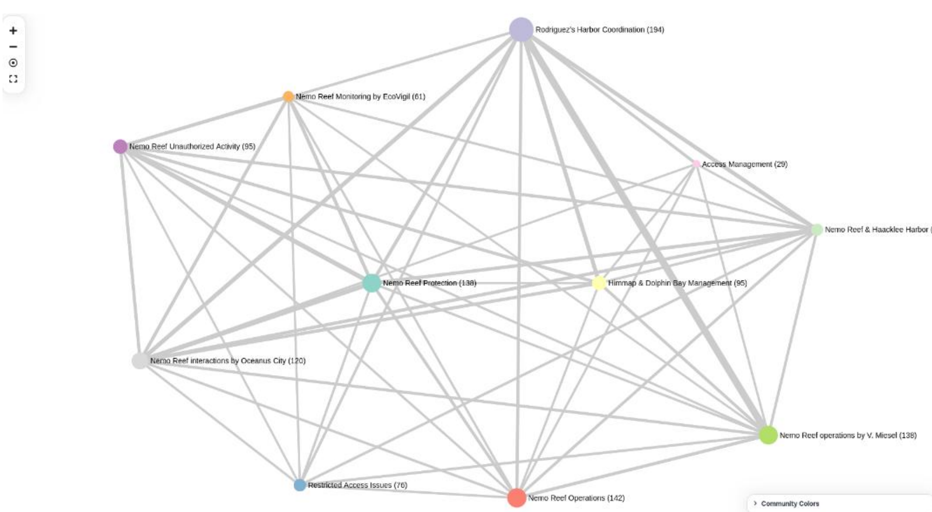


Figure: Community Graph Visualization

With the community data, we can change the colouring of the Timeline as well as the Node Link diagram of the knowledge graph. To help the analysis of specific communities, we can filter the nodes based on the community in the Node Link diagram and.

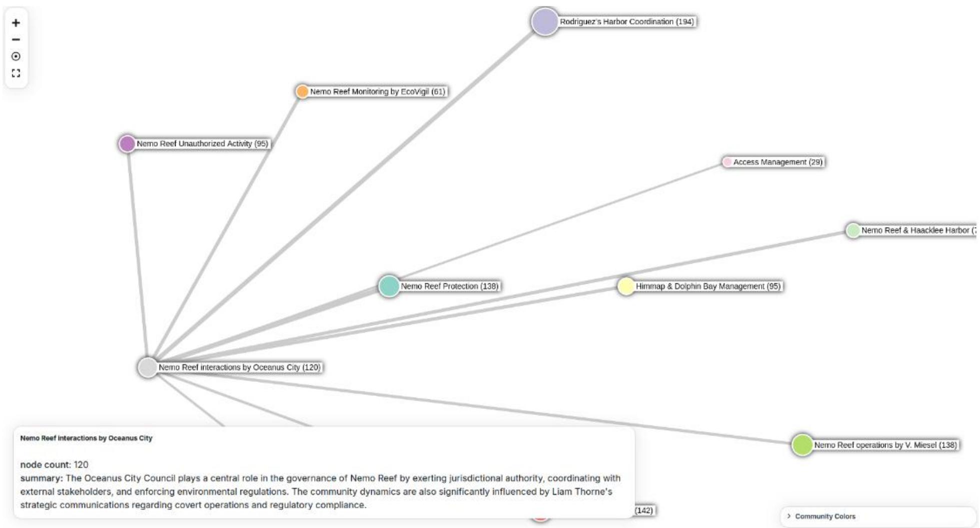


Figure: Community Graph visualization showing the "Nemo Reef interactions by Oceanus City" community highlighted.

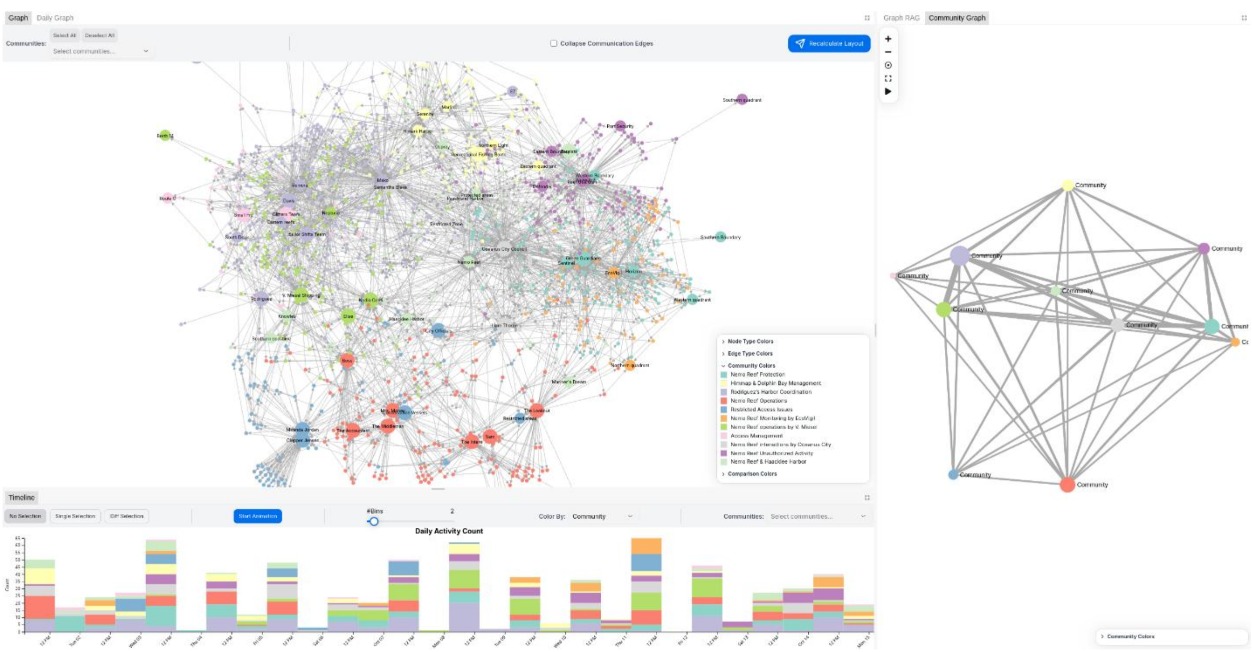


Figure: Knowledge Graph with the coloring based on the community (top) in timeline of events colored by the community (bottom)

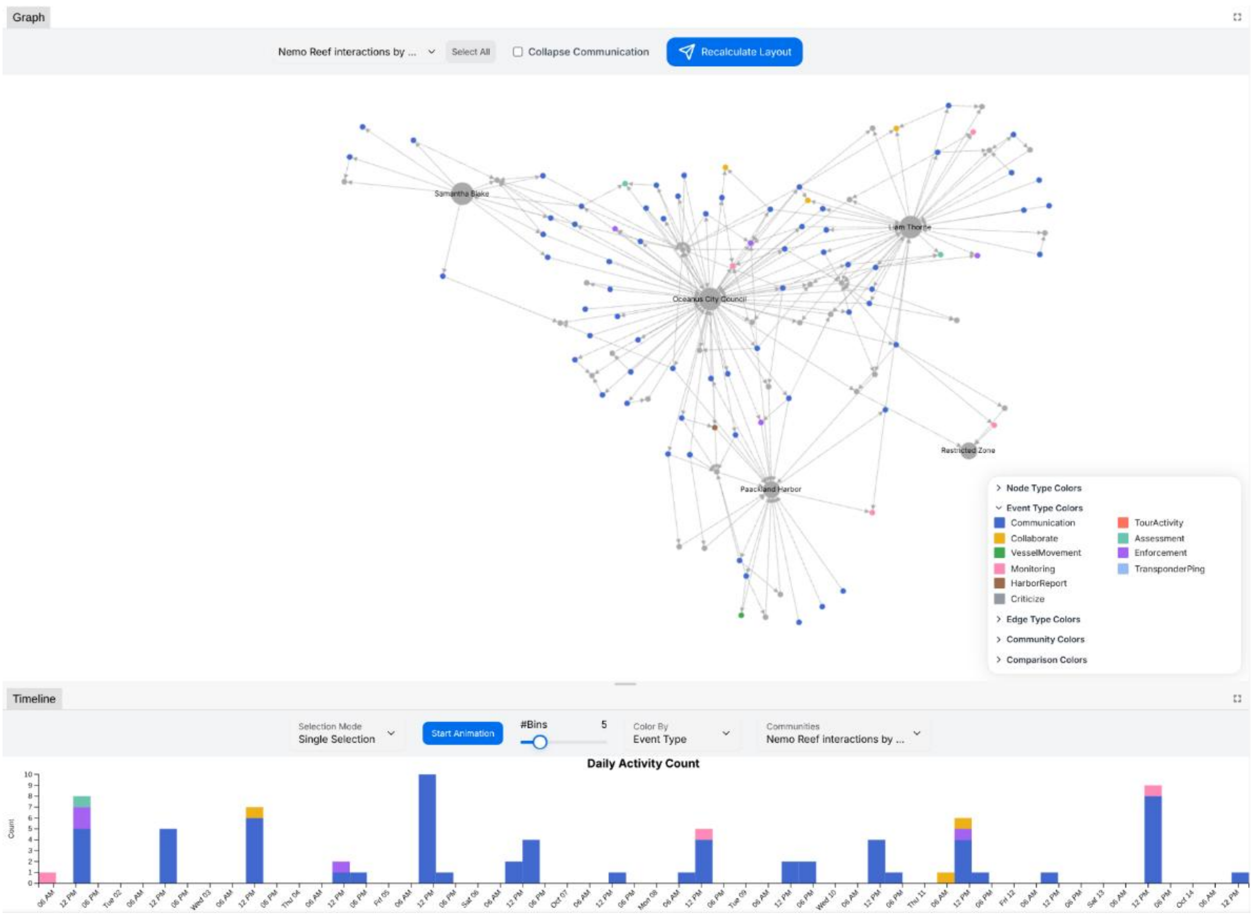


Figure: Knowledge Graph with the community `Nemo Reef interactions by Oceanus City` selected (top) and the timeline of events associated with the same community (bottom).

2b) Community Associations and Predominant Topics

The communities "Nemo Reef Operations by V. Miesel" and "Rodriguez’s Harbor Coordination" are highly interconnected, indicating frequent interactions and possibly shared interests or collaborations. The edge size in the community graph is proportional to the number of interactions between the two communities. For these two communities, the edge size is relatively large.

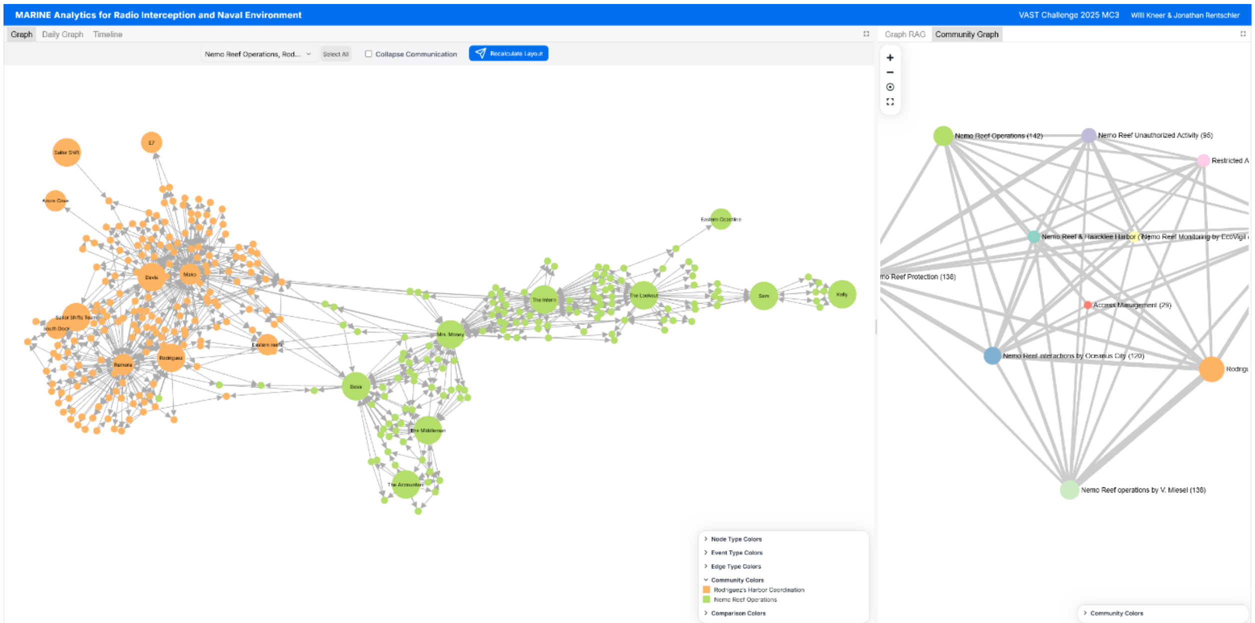


Figure: Community graph with the two communities "Nemo Reef Operations by V. Miesel" and "Rodriguez’s Harbor Coordination" highlighted.

Approximately half of the identified communities are primarily concerned with issues related to Nemo Reef. This highlights that the Nemo Reef is a central topic in the network.

3 – It was noted by Clepper's intern that some people and vessels are using pseudonyms to communicate.

a. Expanding upon your prior visual analytics, determine who is using pseudonyms to communicate, and what these pseudonyms are.
    - Some that Clepper has already identified include: “Boss”, and “The Lookout”, but there appear to be many more.
    - To complicate the matter, pseudonyms may be used by multiple people or vessels.

b. Describe how your visualizations make it easier for Clepper to identify common entities in the knowledge graph.

c. How does your understanding of activities change given your understanding of pseudonyms?

3a) Pseudonym Detection

After looking over the Knowledge Graph on Community of Persons and Names that looks very suspicious:

1. The Accountant (Davis)
2. Boss (Nadia Conti)
3. Small Fry (Rodriguez)
4. The Middleman (Liam Throne)
5. Mrs. Money (Elise)
6. The Intern (Sam)
7. The Lookout (Kelly)

After asking the GraphRAG Pipeline what this Group is all about, we got this Answer:

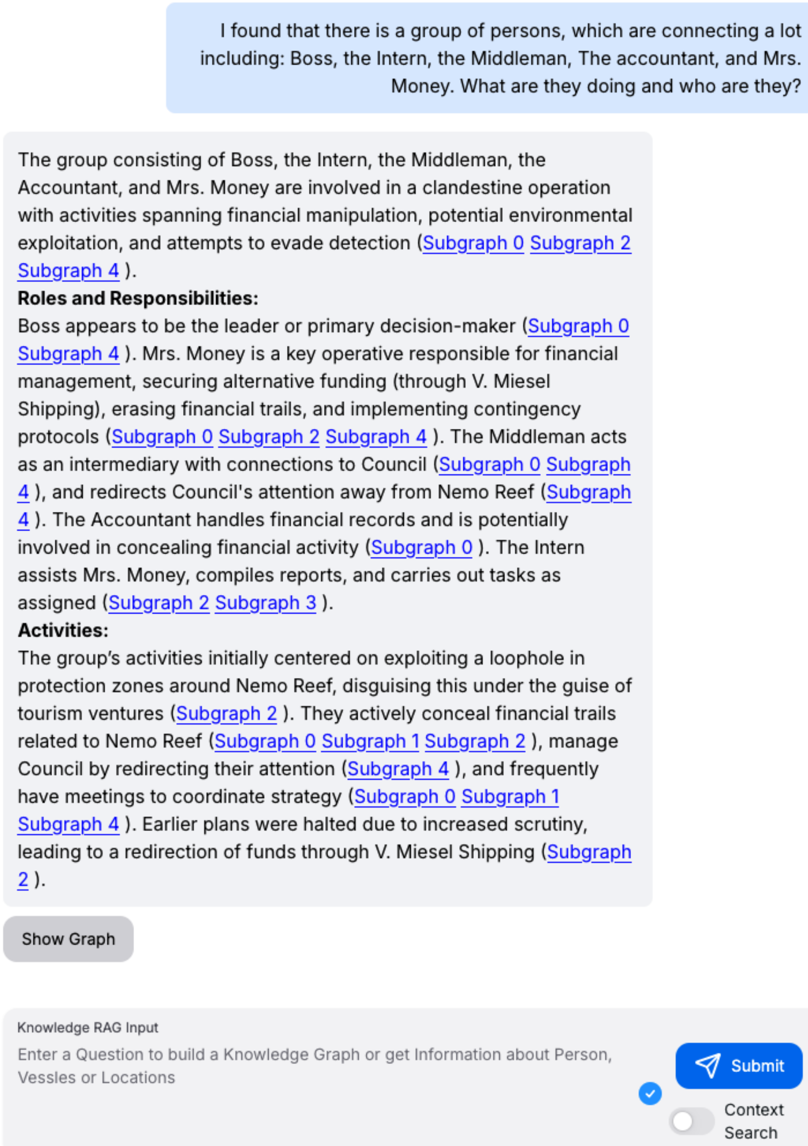


Figure: Prompt "I found that there is a group of persons, which are connecting a lot including: Boss, the Intern, the Middleman, The accountant, and Mrs. Money. What are they doing and who are they?"

The Middleman: Looking at this Information we found that the easiest Entity to uncover would be the Middleman. Because he is a Person talking to the city council, who has specific information only he and the Middleman could have.

So, we selected the Community with the city council in the Bar chart to look at the events they have.

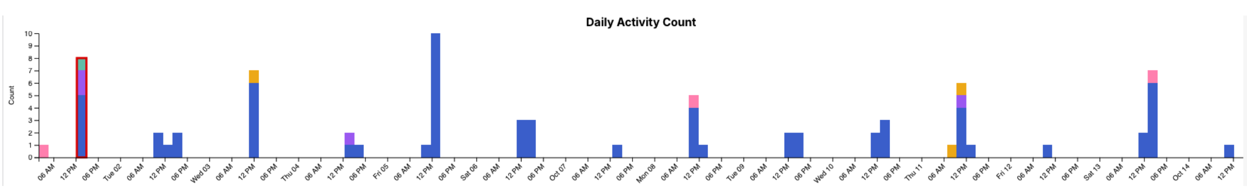


Figure: City Council Events Timeline

And when looking at the events, we also included the Community with all the discovery nick names. Which led to this graph for the first timestamp (01.10 10:00 AM - 01:00 PM).

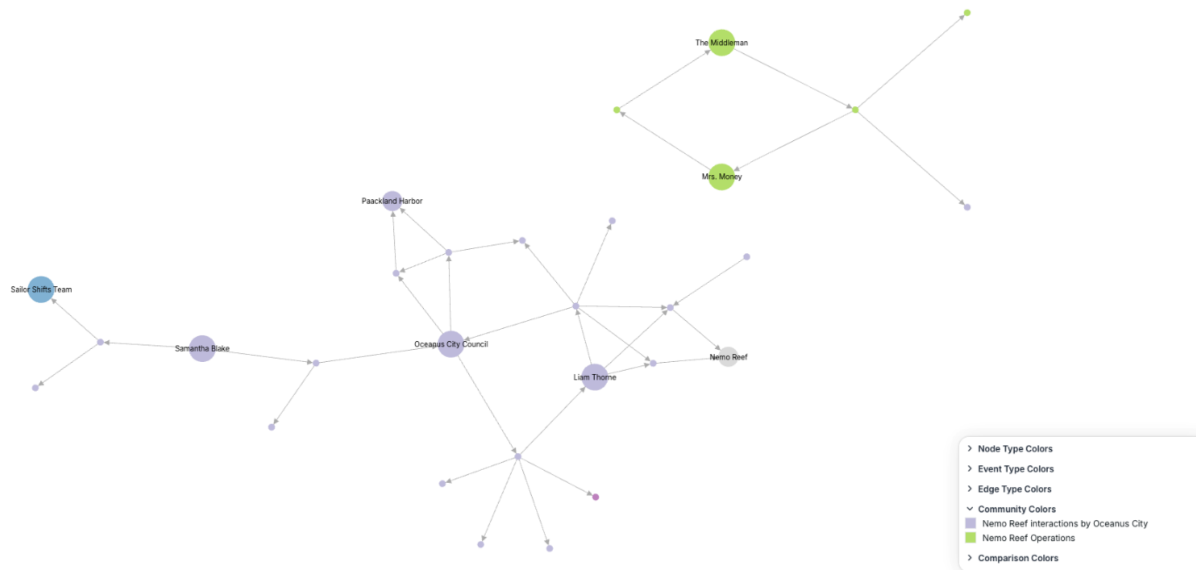


Figure: Liam Thornes @ 01.10 10:00 AM - 01:00 PM

Here at 10:51 AM the City Council asks Liam to increase the patrols at Nemo Reef, because they heard about increased vessel activity there. At 10:53 AM the Middleman sends to Mrs. Money that the Council has flagged increased activity at Nemo Reef.

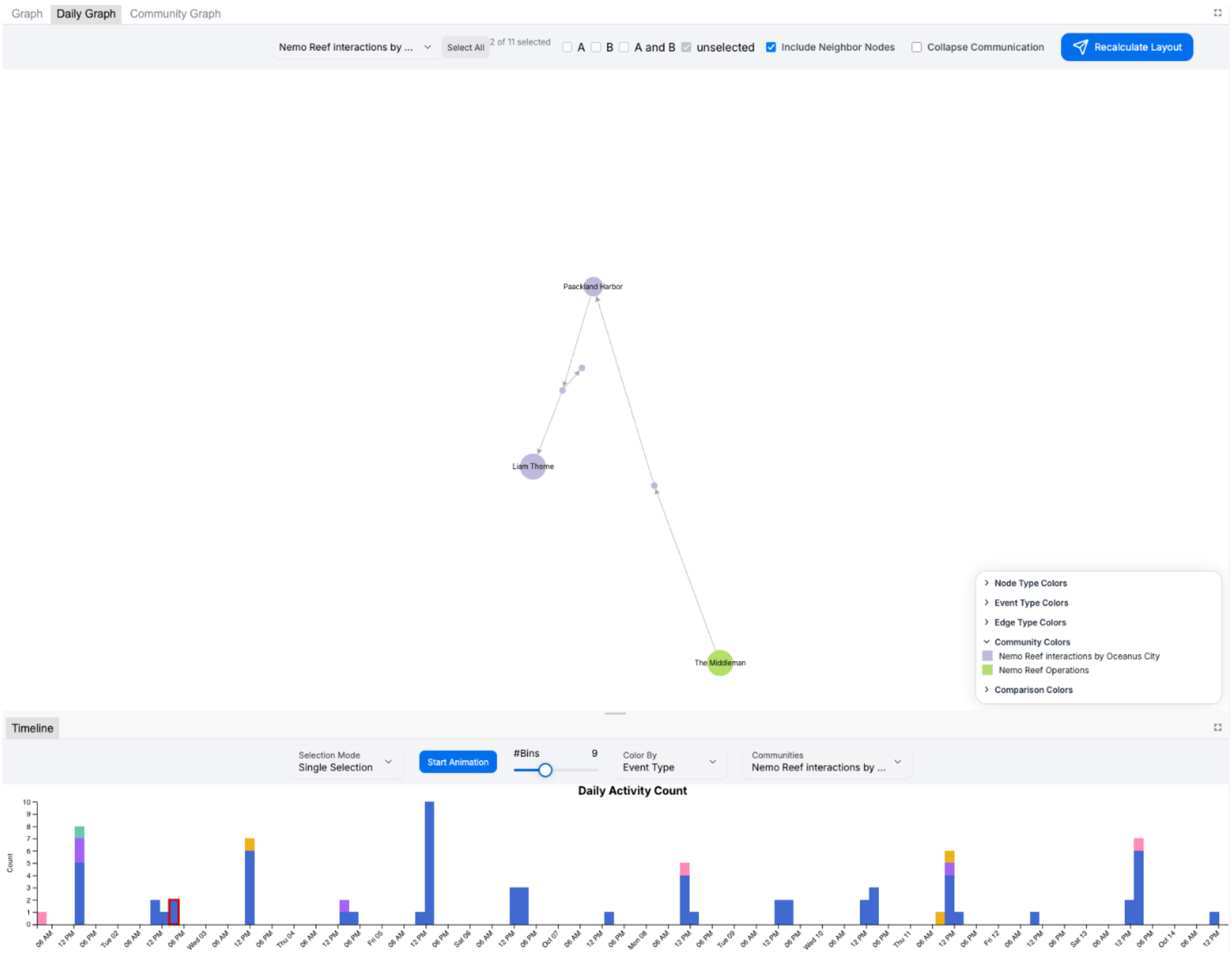


Figure: Liam Thornes as Middleman

And here the Paackland Harbour asks Liam for a report, and he answers as the Middleman. Fully exposing himself.

Mrs. Money: The next easiest Person to uncover was Mrs. Money. While reading the GraphRAG answer, we discovered that Mrs. Money must be a person who often communicates with V. Miesel Shipping. And after taking the same approach as for the Middleman, we found out that Mrs. Money is Ellise.

The Intern and the Lookout: While investigating patterns in the graph, we found that Kelly and Sam's communication and the Lookout's and the Interns' communication are about the same things. With Kelly being the Lookout and Sam being the Intern.

Small Fry: With the GraphRAG we found out that Rodriguez is Small Fry.

3c) Understanding Activities with Pseudonyms

Yes, it did change. Before we knew that Liam Throne was the Middleman, his questions to the Council seemed normal, but after that, you can see the tone shift and how he is seeking information.

4 – Clepper suspects that Nadia Conti, who was formerly entangled in an illegal fishing scheme, may have continued illicit activity within Oceanus.

a. Through visual analytics, provide evidence that Nadia is, or is not, doing something illegal.

b. Summarize Nadia's actions visually. Are Clepper's suspicions justified?

4a) Evidence of Illegal Activity

We can ask the RAG and look at the Subgraphs:

How is Nadia Conti talking to over time?

Nadia Conti's pattern of communication over the specified period reflects her ongoing engagement with colleagues and stakeholders concerning operational strategies and logistical concerns related to Nemo Reef.

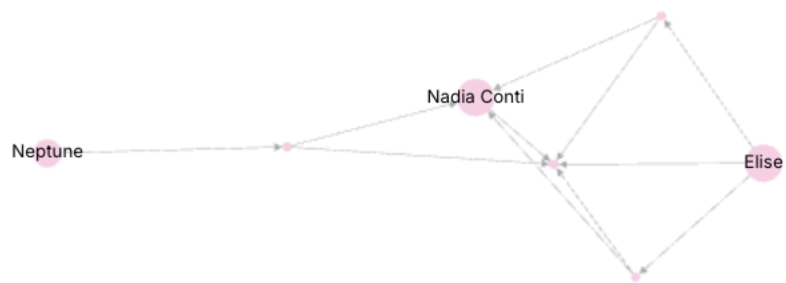
Subgraph 0 (October 8-10, 2040): During this time frame, Nadia's interactions primarily involve Elise and Neptune. The communications focus on preparing for an upcoming meeting at Nemo Reef. On October 8th, critical topics discussed include Sam's insights into shipping lanes intersecting with Mako as well as Neptune's concerns about "underwater foundation work." Furthermore, the discussion touches upon potential adjustments in resource allocation due to an anticipated increase in workload, supported by funding approval from Elise (Subgraph 0). By October 10th, discussions continue with Elise, emphasizing Sam's findings regarding V. Miesel's permanent structures and the impact of increased conservation vessel presence at Nemo Reef. Financial strategies are highlighted as a focal point for ensuring operational adaptability during their meeting (Subgraph 0).

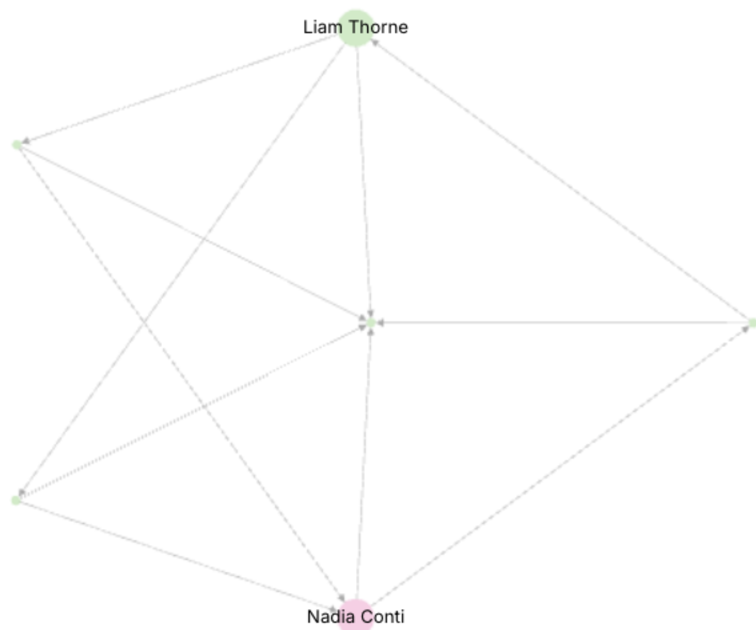
Subgraph 1 (October 10, 2040): This interaction continues with Elise on October 10th, further addressing maritime activities near Nemo Reef and potential scrutiny increases from conservation entities such as EcoVigil. The conversation underlines the necessity of adaptive financial strategies to manage evolving conditions effectively (Subgraph 1).

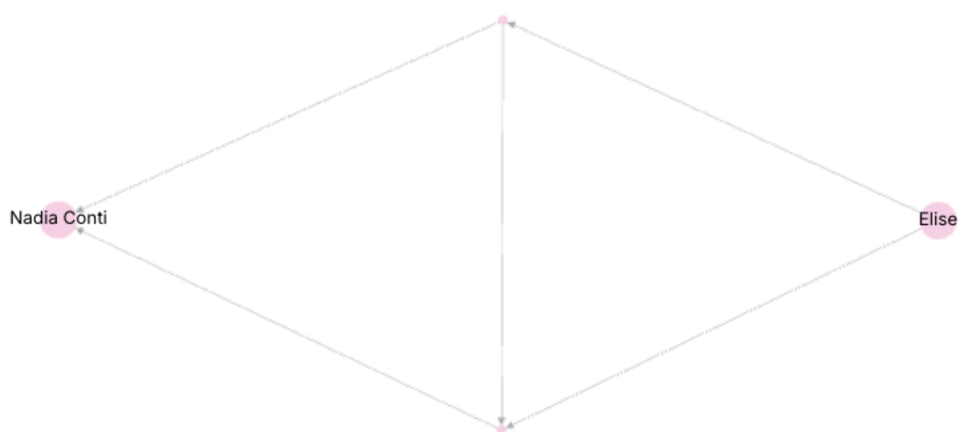
Subgraph 3 (Late October - Early November 2040): This subgraph reveals a transition in Nadia's communications to interactions with Liam Thorne, focusing on operational strategies related to EcoVigil surveillance and clandestine activity evasion near Nemo Reef. The dialogue evolves from organizing covert operations aligned with Harbor Master directives to adjusting operational zones for evading suspicion. These discussions highlight Nadia's dual focus on field operations and diplomatic engagements (Subgraph 3).

Overall, Nadia Conti's communications display a consistent emphasis on strategizing maritime operations, managing resource allocation adaptively, and planning strategically to mitigate environmental and political vulnerabilities related to Nemo Reef. The progression in context and detail across her interactions reflects a dynamic approach towards navigating operational challenges (Subgraph 0 Subgraph 1 Subgraph 3).

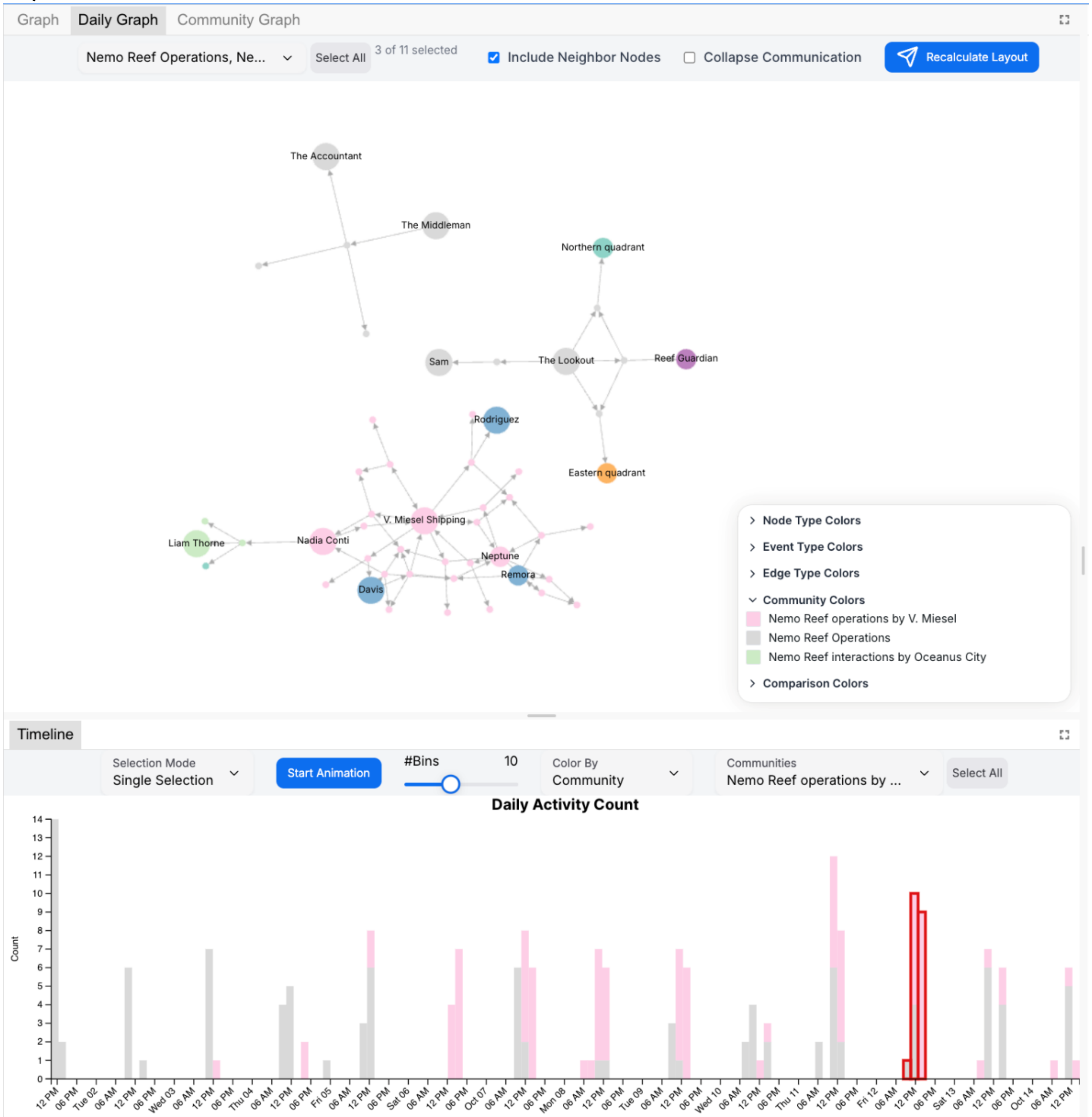
Show Graph


Nadia Conti
Neptune
Elise


Liam Thorne
Nadia Conti


Nadia Conti
Elise

4b)



Here in the Conversation, Nadia is talking to Liam Throne (The Middleman) about a Meeting. And later he talks to The Accountant (Davis) about this Meeting, but not with Nadia, but with the Boss. (Found use RAG-Subgraphs)

Reflection Questions (Optional)

- Given the task to develop visualizations for knowledge graphs, did you find that the challenge pushed you to develop new techniques for visual representation?
- Did you participate in last year’s challenge? If so, did your experience last year help prepare you for this year’s challenge?
- What was the most difficult part of working on this year’s data and what could have made it more accessible?

We did not participate in last year's challenge.

Working with the html Form was a bit difficult, because not all our machines used supported Microsoft Word. Markdown or latex would have been easier to work with.

All VAST data is fully synthetic. Any resemblance to real people, places, or events is purely coincidental. Some elements of the 2025 VAST Challenge resemble data released for the 2024 VAST Challenge. Participants should not assume that there is any continuity and should not use any earlier or any external data for their submission. Mini-Challenge 3 participants should only use data supplied for this mini-challenge in their submission. Data from other challenges should not be combined, except in the Design Challenge.